Learn हिंदी (Hindi) interactively with Sounds

By Vedavyasa Rao

Introduction

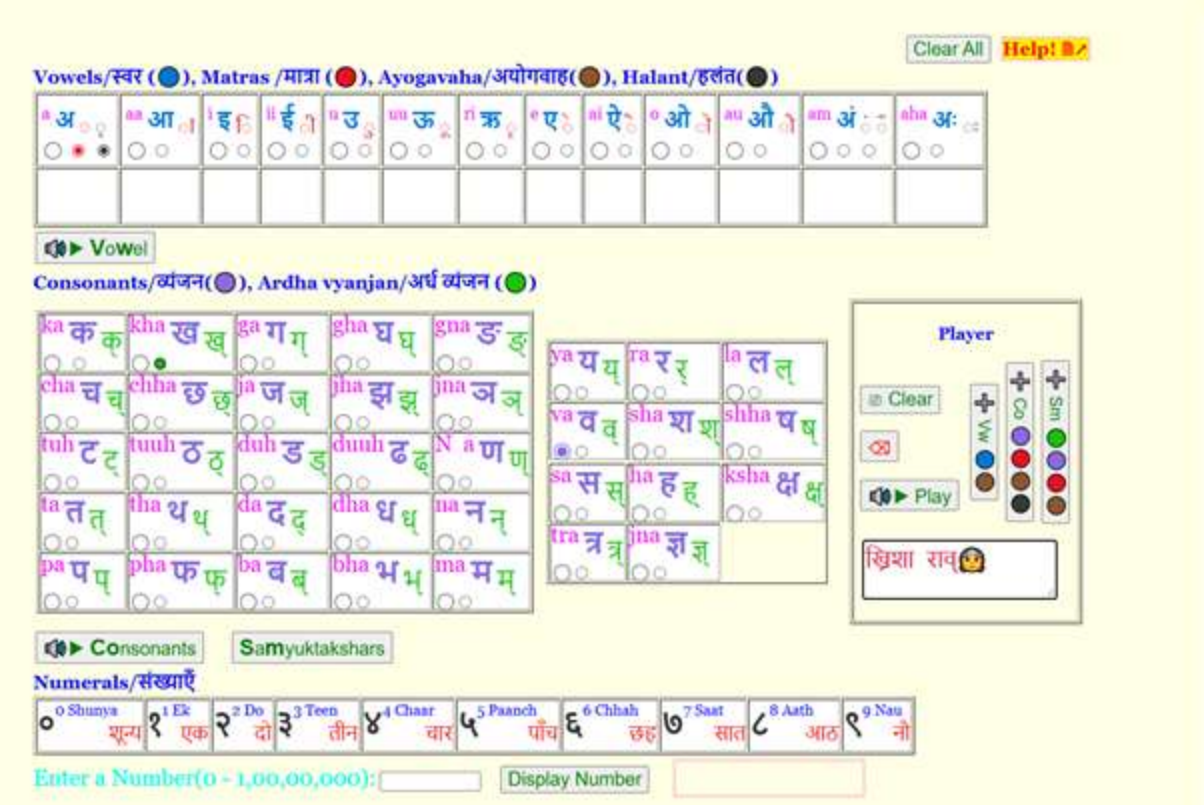
This tool is designed for the audience such as school going kids or adults interested in learning Hindi language by forming words and later hearing the sounds of the words. The sounds are provided by the courtesy of Googles Text to Speech Synthesizers.

The tool comes with preloaded sounds for vowels and consonants with Matras. These are again generated by the courtesy of Googles Text to Speech Synthesizers.

The preloaded sound files are stored and streamed by courtesy of GitHub.

The tool itself is a single HTML page with JavaScript code driving it. Crucial parts of the JavaScript code is generated by the courtesy of Gemini ai.

URL: https://aihinditutor.blogspot.com/



We shall deep dive to understand different aspects of the tool.

Character Groupings

Vowels: अ, आ, इ, ई, उ, ऊ, ऋ, ए, ऐ, ओ, औ, अं, अः

Matras: ◌, ◌ा, ◌ि, ◌ी, ◌ु, ◌ू, ◌ृ, ◌े, ◌ै, ◌ो, ◌ौ

Halant: ◌्

Ayogavaha: ◌ं,◌ँ, ◌ः

Consonants:

क, ख, ग, घ, ङ,

च, छ, ज, झ, ञ,

ट, ठ, ड, ढ, ण,

त, थ, द, ध, न,

प, फ, ब, भ, म,

य, र, ल, व, श, ष, स, ह, क्ष, त्र, ज्ञ

Ardha vyanjans:

क्, ख्, ग्, घ, ङ्,

च्, छ्, ज्, झ्, ञ्,

ट्, ठ्, ड्, ढ्, ण्,

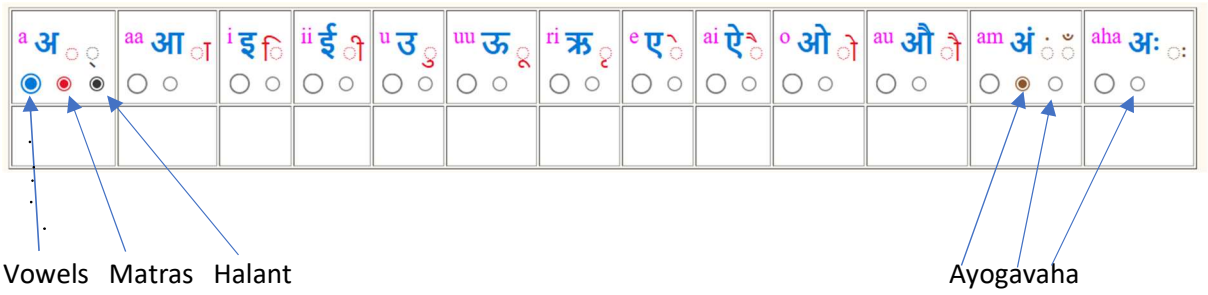
त्, थ्, द्, ध्, न्,

प्, फ्, ब्, भ्, म्,

य्, र्, ल्, व्, श्, ष्, स्, ह्, क्ष्, त्र्, ज्ञ्, श्र्

Radio button mappings

Vowels, Matras, Halant and Ayogavaha



Vowels Matras Halant Ayogavaha

(Anusvara, Chandrabindu, Visarga)

Consonants and Ardha vyanjans



Consonant Ardha vyanjan

Notice that the radio buttons have matching colors as the text.

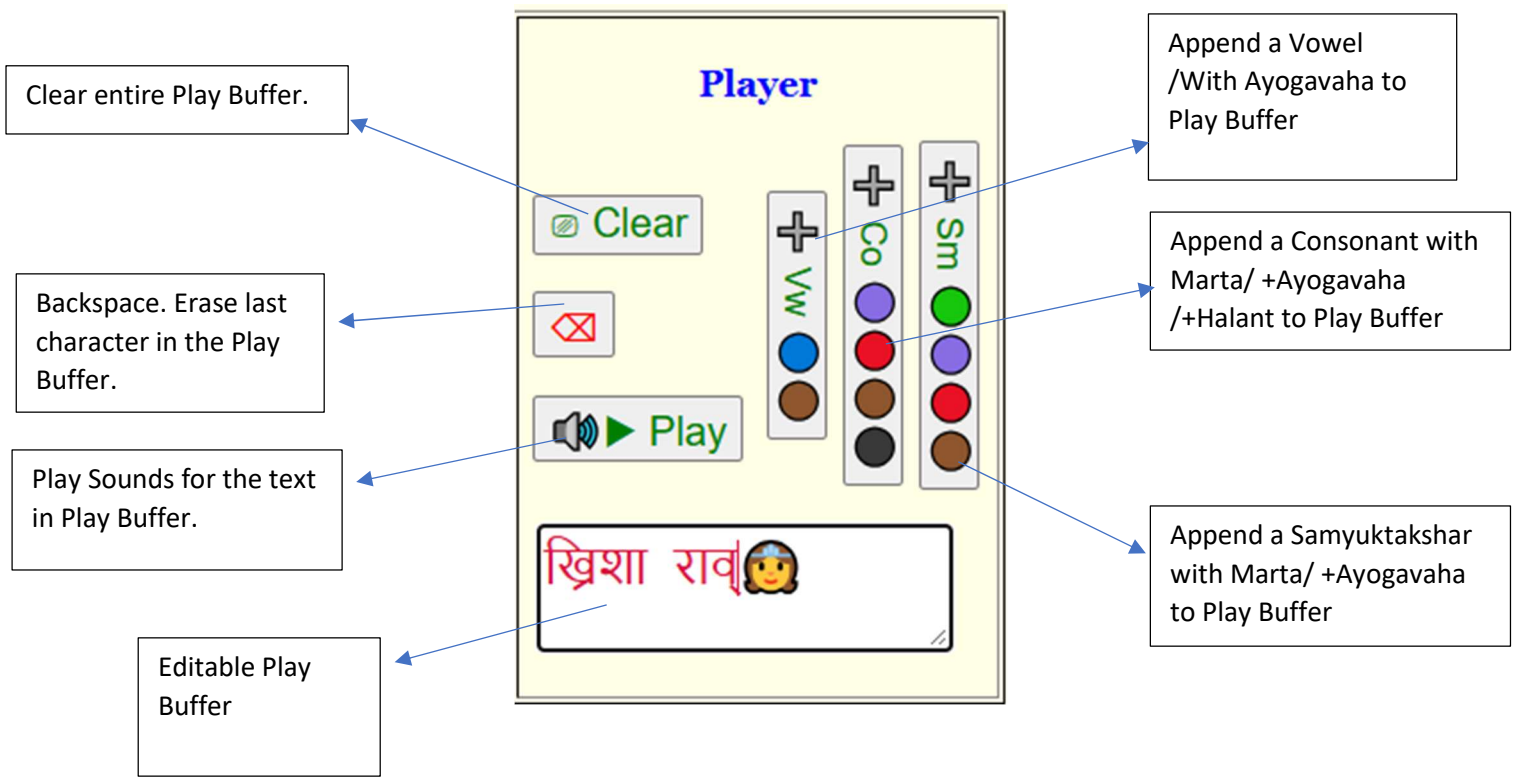
Clear All

Clicking this button clears all Radio button selections.

Help!

Displays this PDF file.

Player



Grammar Rules

For Vowels, Vowels (🔵) and Ayogavaha (🟤) are allowed. Examples, ई, ऊँ

For Consonants, Consonants (🟣), Matras (🔴), Halant (⚫), Ayogavaha (🟤), and are allowed. Examples, राम् शंकर

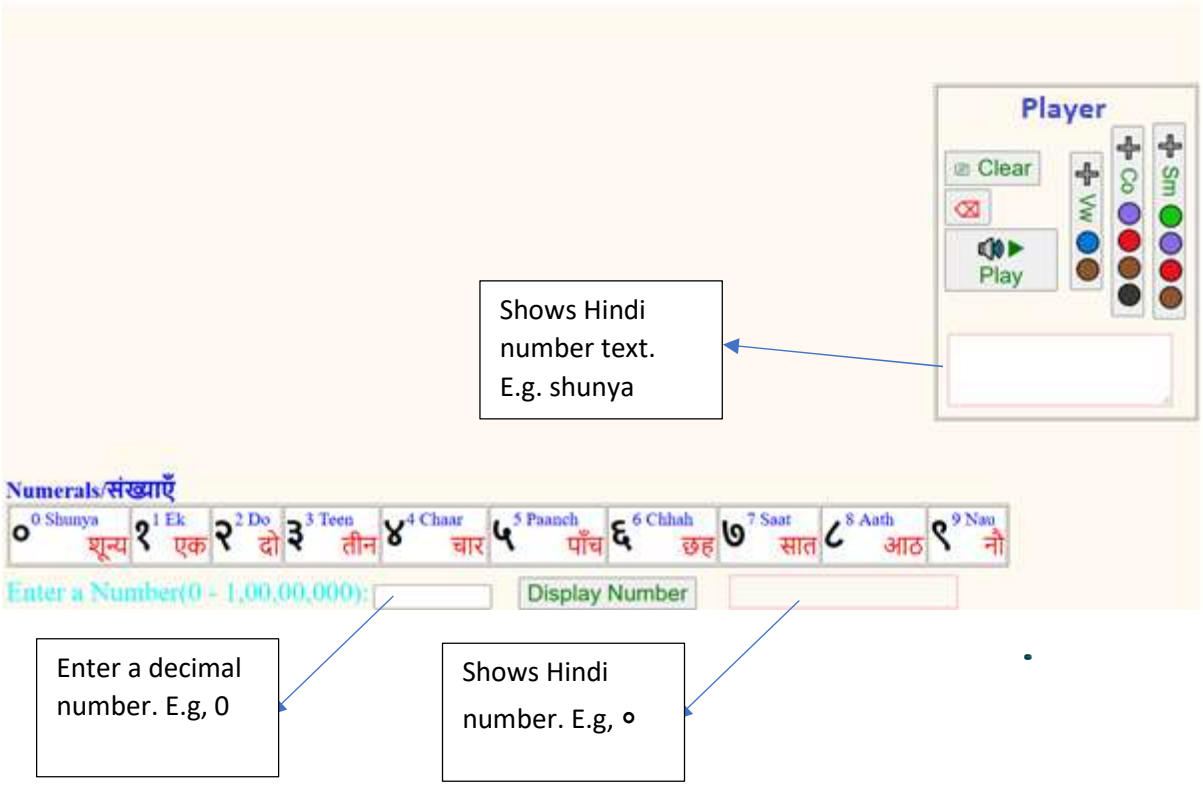
For Samyuktakshars, Ardha vyanjans (🟢), Consonants(🟣), Ayogavaha (🟤), and Matras (🔴) are allowed. Examples, श्याँ

The player is editable, so you can separate letters by putting comma. This will slow down the playback.

For e.g., क्र,क्रा,क्रि,क्री,क्रु,क्रू,क्रृ,क्रे,क्रै,क्रो,क्रौ,क्रं,क्रः

Numerals

The numbers can be entered into the input box. Clicking the “Display Button” prints the number and populates its hindi text in the “Play Buffer” for audio sound as shown below.

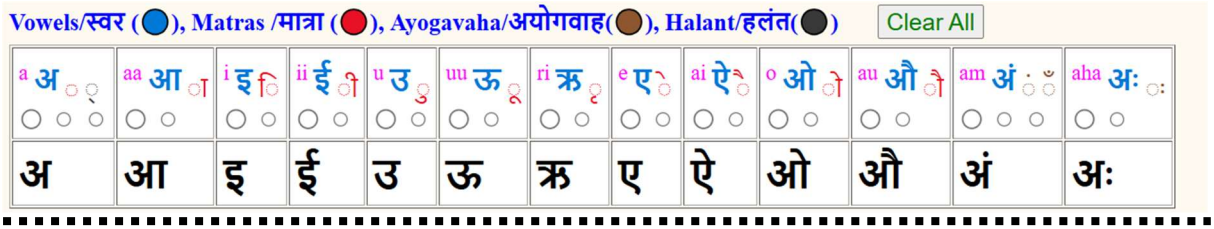


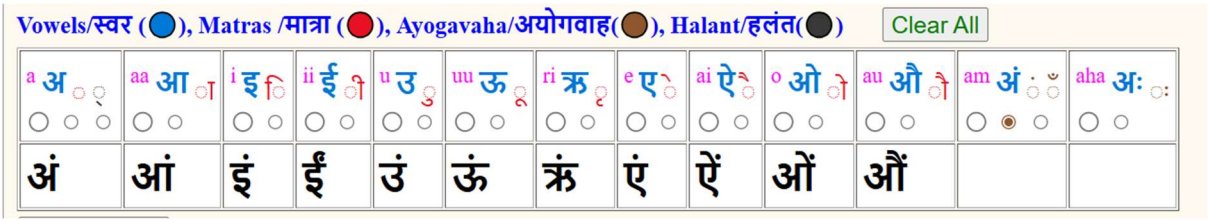
Use cases

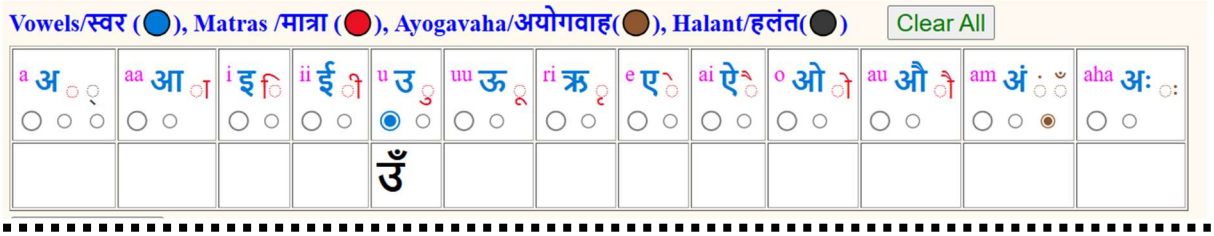


Prints Vowels and Plays associated sound.

- [x] Vowels (🔵)
- [ ] Consonants(🟣)
- [ ] Ardha vyanjans (🟢)
- [x] Ayogavaha (🟤)
- [ ] Matras (🔴)
- [ ] Halant (⚫)









Prints Consonants and Plays associated sound.

- ✕ Vowels (🔵)
- ☑ Consonants(🟣)
- ✕ Ardha vyanjans (🟢)
- ☑ Ayogavaha (🟤)
- ☑ Matras (🔴)
- ☑ Halant (⚫)







Samyuktakshars

Prints Samyuktakshars only. The sounds can be generated from the player by separating generated text with a comma as shown below.

For e.g., क्र,क्रा,क्रि,क्री,क्रु,क्रू,क्रृ,क्रे,क्रै,क्रो,क्रौ,क्रं,क्रः

✕ Vowels (🔵)

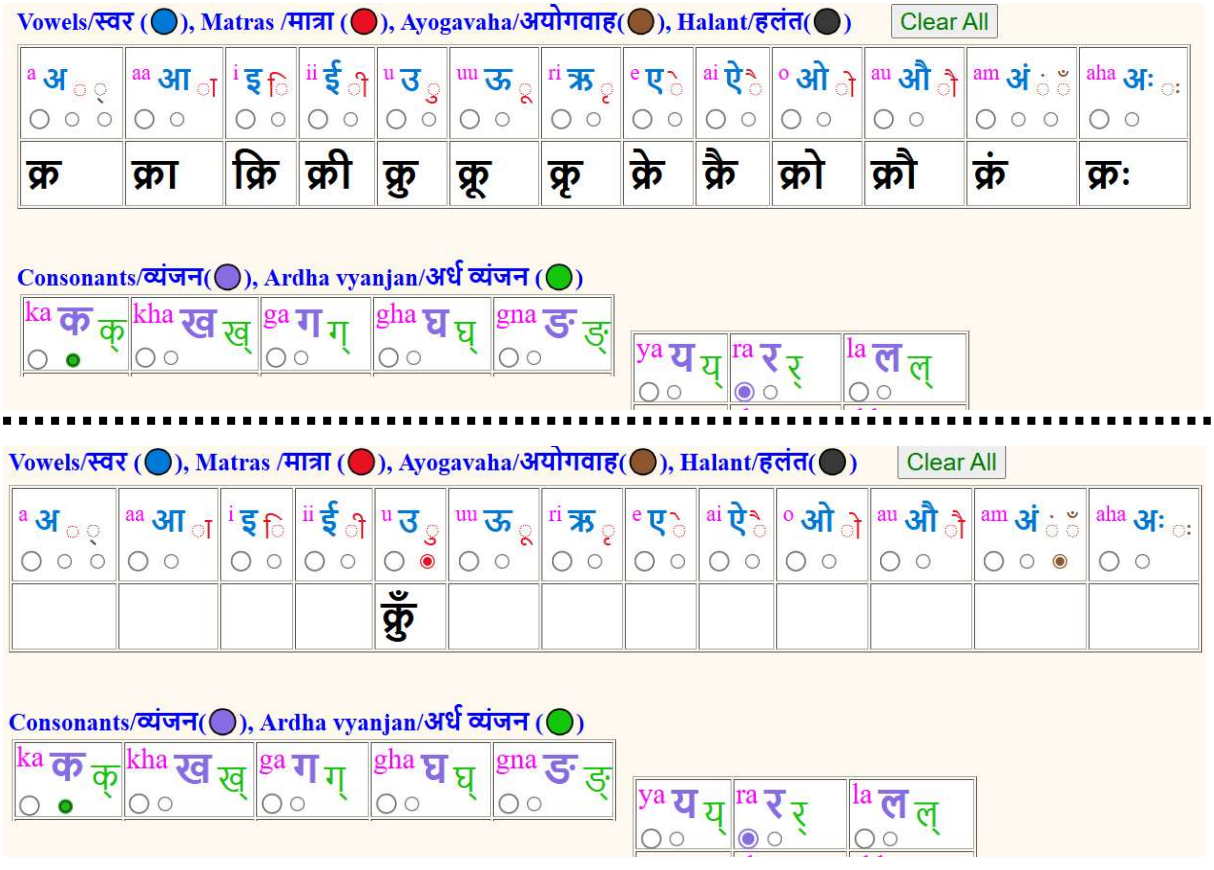
☑ Consonants(🟣)

☑ Ardha vyanjans (🟢)

☑ Ayogavaha (🟤)

☑ Matras (🔴)

✕ Halant (⚫)



Vowels/स्वर (🔵), Matras /मात्रा (🔴), Ayogavaha/अयोगवाह(🟤), Halant/हलंत(⚫)

a अ	aa आ	i इ ि	ii ई ी	u उ ु	uu ऊ ू	ri ऋ ृ	e ए े	ai ऐ ै	o ओ ो	au औ ौ	am अं ं	aha अः ः
क्र	क्रा	क्रि	क्री	क्रु	क्रू	क्रृ	क्रे	क्रै	क्रो	क्रौ	क्रं	क्रः

Consonants/व्यंजन(🟣), Ardha vyanjan/अर्ध व्यंजन (🟢)

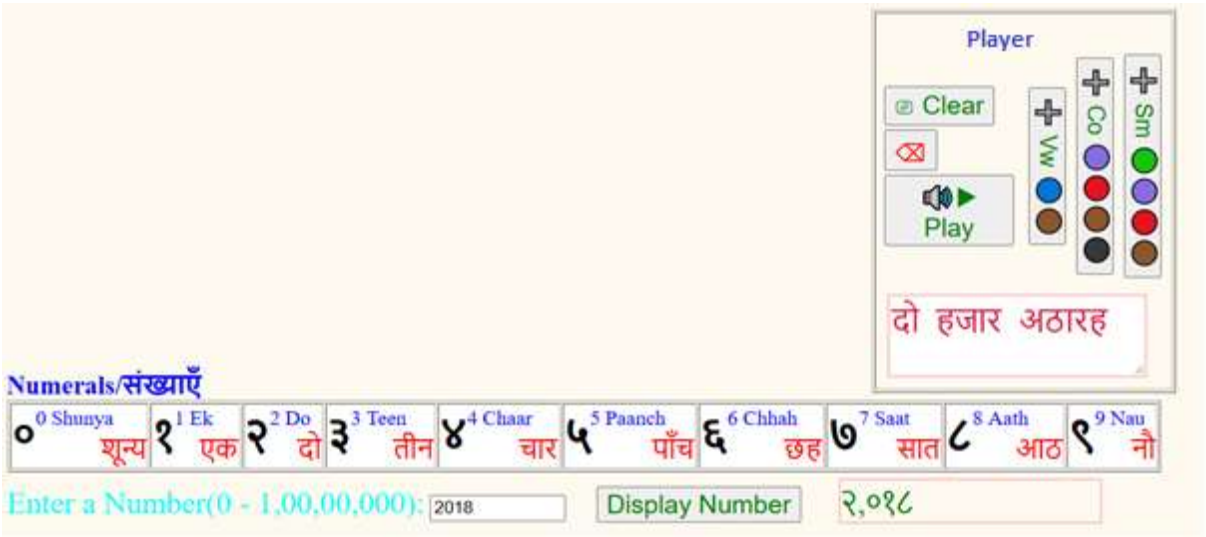
ka क क्	kha ख ख्	ga ग ग्	gha घ घ्	gna ङ ङ्

ya य य्	ra र र्	la ल ल्



Display Number

Converts decimal number into Hindi number and hindi text for the same.



Troubleshooting and workarounds

Preloaded Sounds

Sometimes the Audio files don't get loaded for streaming. In such cases, the Vowel or Consonants button background color becomes red as shown below.



First make sure radio button selections are made to play sounds.

You can retry by clicking on the red button several times till it becomes green when reloading becoming successful.

If it still stays red, reload the page in the browser.

If it still stays red, you can continue and use player for sounds.

Sound Quality on Mobile devices

Due to CORS network restrictions, locally installed device voices are used instead of high-quality Google neural models. For the best sound quality and experience, please use a Desktop or Laptop if possible.

Reach out!

If you observe any inaccuracies or inconsistencies, kindly drop me an email at rvvyas@gmail.com with the subject starting with "aihinditutor".